



SQL & DBMS Practice Questions

Sample Tables

DEPARTMENT

```
CREATE TABLE DEPARTMENT (  
    dept_id INTEGER PRIMARY KEY,  
    dept_name TEXT NOT NULL,  
    location TEXT  
);  
  
INSERT INTO DEPARTMENT VALUES (1, 'Sales', 'Chennai');  
INSERT INTO DEPARTMENT VALUES (2, 'Accounting', 'Mumbai');  
INSERT INTO DEPARTMENT VALUES (3, 'IT', 'Bangalore');  
INSERT INTO DEPARTMENT VALUES (4, 'HR', 'Hyderabad');  
INSERT INTO DEPARTMENT VALUES (5, 'Marketing', 'Delhi');
```

EMPLOYEE

```
CREATE TABLE EMPLOYEE (  
    empId INTEGER PRIMARY KEY,  
    name TEXT NOT NULL,  
    dept_id INTEGER,  
    salary INTEGER,  
    hire_date DATE  
);  
  
INSERT INTO EMPLOYEE VALUES (1, 'Clark', 1, 50000, '2022-01-10');  
INSERT INTO EMPLOYEE VALUES (2, 'Dave', 2, 60000, '2021-03-15');  
INSERT INTO EMPLOYEE VALUES (3, 'Ava', 1, 55000, '2023-07-01');  
INSERT INTO EMPLOYEE VALUES (4, 'John', 3, 70000, '2020-05-20');  
INSERT INTO EMPLOYEE VALUES (5, 'Sara', 3, 75000, '2019-11-11');  
INSERT INTO EMPLOYEE VALUES (6, 'Mike', 4, 48000, '2021-09-01');  
INSERT INTO EMPLOYEE VALUES (7, 'Nina', 5, 52000, '2022-02-14');  
INSERT INTO EMPLOYEE VALUES (8, 'Liam', 3, 72000, '2020-12-20');  
INSERT INTO EMPLOYEE VALUES (9, 'Emma', 2, 61000, '2023-01-05');  
INSERT INTO EMPLOYEE VALUES (10, 'Olivia', 1, 53000, '2022-05-17');
```

PROJECT

```
CREATE TABLE PROJECT (  
    project_id INTEGER PRIMARY KEY,  
    project_name TEXT NOT NULL,  
    dept_id INTEGER  
);  
  
INSERT INTO PROJECT VALUES (101, 'Project Alpha', 1);  
INSERT INTO PROJECT VALUES (102, 'Project Beta', 2);  
INSERT INTO PROJECT VALUES (103, 'Project Gamma', 3);  
INSERT INTO PROJECT VALUES (104, 'Project Delta', 4);  
INSERT INTO PROJECT VALUES (105, 'Project Epsilon', 5);  
INSERT INTO PROJECT VALUES (106, 'Project Zeta', 1);  
INSERT INTO PROJECT VALUES (107, 'Project Theta', 3);
```

EMP_PROJECT

```
CREATE TABLE EMP_PROJECT (  
    empId INTEGER,  
    project_id INTEGER,  
    hours_worked INTEGER,  
    PRIMARY KEY (empId, project_id)  
);  
  
INSERT INTO EMP_PROJECT VALUES (1, 101, 40);  
INSERT INTO EMP_PROJECT VALUES (1, 106, 20);  
INSERT INTO EMP_PROJECT VALUES (3, 101, 30);  
INSERT INTO EMP_PROJECT VALUES (3, 106, 25);  
INSERT INTO EMP_PROJECT VALUES (2, 102, 25);  
INSERT INTO EMP_PROJECT VALUES (2, 105, 15);  
INSERT INTO EMP_PROJECT VALUES (4, 103, 50);  
INSERT INTO EMP_PROJECT VALUES (4, 107, 30);  
INSERT INTO EMP_PROJECT VALUES (5, 103, 60);  
INSERT INTO EMP_PROJECT VALUES (8, 107, 40);  
INSERT INTO EMP_PROJECT VALUES (6, 104, 35);  
INSERT INTO EMP_PROJECT VALUES (7, 105, 45);  
INSERT INTO EMP_PROJECT VALUES (9, 102, 20);  
INSERT INTO EMP_PROJECT VALUES (10, 101, 15);
```

SQL Questions – 50 Questions

Section 1 : Basics (10 Questions)

- 1 . Select all columns from EMPLOYEE .
- 2 . Select only employee names from EMPLOYEE .
- 3 . Select employees with salary > 60,000 .
- 4 . Count total employees in each department.
- 5 . Find the average salary in the company.
- 6 . Find the highest salary in each department.
- 7 . List distinct departments from EMPLOYEE .
- 8 . Order employees by salary descending.
- 9 . Select employees hired after 01-01-2022 .
- 10 . Find employees in the 'Sales' department.

Section 2 : Intermediate (10 Questions)

- 1 . Find the second highest salary in EMPLOYEE .
- 2 . Find employees earning more than the average salary of their department.
- 3 . List employees along with department name using JOIN.
- 4 . Find employees not assigned to any project.
- 5 . Count number of employees working on multiple projects.
- 6 . Total hours worked per project.
- 7 . Employees working on projects of their own department.
- 8 . Departments having more than 1 employee.
- 9 . Find employees hired in the same month.
- 10 . Employees with salaries between 50,000 and 70,000 .

Section 3 : Advanced (10 Questions)

- 1 . Top 2 highest paid employees in each department.
- 2 . Find employees with the second highest salary per department.
- 3 . Find projects with total hours exceeding 50 .
- 4 . List employees who have never worked on any project.
- 5 . Monthly count of employees hired per department.
- 6 . Employees with tenure greater than average tenure.
- 7 . Employees working on all projects.
- 8 . Departments with no employees.
- 9 . Employees with salary greater than their manager.
- 10 . Top 3 projects by hours worked per department.

Section 4 : Date & Time (10 Questions)

- 1 . Find employees hired in the last 6 months.
- 2 . Find employees with hire anniversaries this week.

- 3 . Days each employee has been in the company.
- 4 . Employees hired in a specific month (e.g., January).
- 5 . Count employees hired per year.
- 6 . Employees hired in the last 30 days.
- 7 . Find employees whose tenure is above average.
- 8 . List employees hired before 2021.
- 9 . Employees hired in the same month as another employee.
- 10 . Employees whose hire_date is a weekend.

Section 5 : Window Functions & Advanced Queries (10 Questions)

- 1 . Use ROW_NUMBER to list top 3 salaries per department.
- 2 . Use DENSE_RANK to list second highest salary per department.
- 3 . Find cumulative sum of hours worked per project.
- 4 . Rank employees by salary within their department.
- 5 . Find employees whose salary is above the median salary.
- 6 . Top N employees by project hours worked.
- 7 . Employees working on more than 2 projects.
- 8 . Find department-wise average salary ranking.
- 9 . Employees with salary in top 10 % of the company.
- 10 . Calculate total hours worked per department across all projects.